

When IEEE 802.11 and 5G Meet Time-Sensitive Networking

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ABSTRACT Many emerging applications require a higher level of flexibility, modularity, and efficiency but are dependent on advancements in communication infrastructure and distributed computing. Time-sensitive networking (TSN) standards aim at providing vendor agnostic, reliable, and deterministic communications over the Ethernet, but lack in flexibility and modularity provisions. In this context wireless communication systems are preferred given the obvious benefits in terms of increased flexibility, reduced deployment & maintenance costs, and inherent mobility support. However, the stochastic nature of the wireless medium poses several challenges in achieving these benefits. In this paper we comprehensively analyze the recent standardization efforts and developments in IEEE 802.11 and 5G to enable low-latency, deterministic communications and present the current status of their integration with wired TSN. Then, we present a set of use cases that may be enabled by wireless TSN including industrial automation, automotive, or audiovisual applications.

INDEX TERMS Real time, deterministic, reliable, time sensitive networking, TSN, 5G, Wi-Fi, IEEE 802.11.

I. INTRODUCTION

The development of future information technology (IT) and operational technology (OT) is being promoted by the vision of flexible, enhanced intelligence, and modular networks. The communication requirements for various applications are both demanding and diverse [1]. As an example, the growing complexity in industrial sensing and closed-loop control systems require the use of heterogeneous sensors, actuators, and control units that have to operate in a predictable and synchronous manner. The elements of the control system can be distributed over short or long distances, and thus the use of communication networks is compulsory to obtain a cost-effective solution. Additionally, there is a high demand on the degree of flexibility provided by the communication network given the increasing number of mobile devices. On the one hand, the wired communication solutions that provide deterministic latency and high reliability may require cumbersome and expensive cabling or may not be even suitable due to lack of functionality such as mobility [2], [3]. On the other hand,

the current state of wireless solutions do not provide the deterministic latency and high reliability required for emerging use cases.

The time-sensitive networking (TSN) task group (TG) [4] is working on a set of standards that enable vendor agnostic technology for reliable and deterministic communications over Ethernet. TSN provides mechanisms like time aware shaper (TAS) [5] and frame preemption [6] to enable co-existence of different traffic types on the same network. Even though TSN is gaining attention due to its advanced features and capabilities to fulfill the quality of service (QoS) of industrial applications, it still lacks the flexibility of wireless networks required by the numerous emerging industrial use cases such as industrial robots, remote assistance, and human machine interface (HMI) [1].

Extending the capabilities of wired TSN network to the wireless domain (wireless TSN) would enable reconfigurable and adaptable communication systems, reduce installation and maintenance costs [7], [8], and support future use cases

involving mobile devices. We define the term “wireless TSN” as a wireless technology that provides a QoS equivalent to the TSN set of standards in terms of reliability, deterministic communication (bounded latency) and time synchronization and that supports the configuration of the networks’ QoS either via integration with wired technology (Ethernet) or as an independent wireless system. The main technologies to achieve wireless TSN capabilities that also have the potential to drive the next generation of industrial wireless communications are IEEE 802.11 (named in the paper as “802.11” and commercially labeled as Wi-Fi) and 3rd generation partnership project (3GPP) fifth generation (5G) cellular system.

On the one hand, the IEEE 802.11 communication standard [9] is continuously evolving to support very demanding applications in different sectors, such as factory automation, healthcare, and in-vehicle networking. One of the main focus of 802.11ax (Wi-Fi 6) was to enhance the communication efficiency and throughput. The upcoming release 802.11be (Wi-Fi 7) is expected to provide guaranteed low-latency and high-reliability over unlicensed bands. To do so, several new mechanisms are being proposed for the new version of the standard, as the scheduled trigger frame (TF)-based access scheme. It is expected that these new features will be key to enable 802.11-based TSN communications.

On the other hand, and similar to 802.11, the 5G communication standard also supports wireless communication for multiple domains including transportation, healthcare, agriculture, and Industry 4.0. The low latency [10] and high reliability provided through ultra-reliable low latency communication (URLLC) enabling technologies support TSN integration with 5G [11]. The 5G system (5GS) acts as a logical bridge to support integration with the TSN standards. As specified in 3GPP Release-16, 5G supports most TSN standards through interaction with TSN application function (AF), network-side TSN translator (NW-TT), and device-side TSN translator (DS-TT) functions. All other core and radio access network (RAN) functionalities remain unexposed to the TSN domain. The TSN AF interacts with the TSN domain for controlling and configuration management, NW-TT interacts with the TSN domain to support connectivity with the 5GS (exchange of data and control information), and DS-TT interacts with the end stations such as the sensors, controllers, and actuators to enable time-sensitive wireless communication. Further enhancements on 5G TSN integration are also expected in the forthcoming 3GPP Release-17.

The concept to enable TSN over the wireless medium has been brought few years ago. A few available studies show how to enable TSN capabilities over the wireless physical (PHY) layer for industrial use cases. The research challenges and gaps for wireless TSN along with available enhancements in 802.11 networks to support TSN capabilities are highlighted in [12]. In particular expected enhancements of 802.11be for enabling TSN capabilities are discussed in [13]. Similarly, to enable TSN over the 5GS, challenges to achieve end-to-end time synchronization and scheduling are presented in [14]. Authors in [15] discuss different enhancements and ongoing

standardization activities related to 5G TSN integration. A roadmap to integrate wireless TSN into wired TSN without a particular focus on any wireless technology is also provided [16]. Additionally, literature is also available on both 802.11 and 5G to extend TSN capabilities over wireless technologies [17], [18].

Nevertheless, some shortcomings in the mentioned literature can be observed. The available literature lacks in providing a comprehensive analysis of the current state of technologies and does not develop deep details on the TSN integration with 802.11 and 5G. Consequently, this paper provides details on the current existing mechanisms of both 802.11 and 5G to support TSN-like communications, and their integration with wired TSN, including: how time synchronization can be achieved between 802.11 or 5G and TSN devices, which techniques can be used to achieve reliability requirements, how TSN traffic priorities will be translated for 802.11 or 5G, and how QoS profiles can be mapped with the TSN defined traffic, etc. Additionally, we present an up-to-date information on techniques and enhancements for 802.11 and 5G with the latest releases of the 802.11 standards and 3GPP specifications that is missing in existing literature.

The paper is organized as follows. Section II as a background, provides an overview of existing wireless communication solutions for industrial use cases. Section III presents an overview of wired TSN standards. In Section IV the challenges for enabling wireless TSN are presented. The mechanisms to enable wireless TSN using 802.11 technology are detailed in Section V. Afterwards, we present the mechanisms to enable wireless TSN using 5G technology in Section VI. Different wireless TSN use cases along with their QoS requirements are presented in Section VII. Finally, we conclude along with the next steps in Section VIII.

II. BACKGROUND

The communication technology landscape for applications requiring deterministic communication is highly dominated by proprietary or standardized but not interoperable wired communication systems. Examples include PROFIBUS [19], EtherCAT [20], SERCOS III [21], and many more [1], [22]. A clear downside of current industrial fieldbuses for deterministic communications is that the solutions are not interoperable and cannot coexist in the same network. To this end, the IEEE 802.1 TSN TG has been working to establish a set of standards to provide deterministic and more reliable communications over standard Ethernet [4]. This effort is also mainly aimed to enable seamless interoperability among different industrial devices, enable their coexistence over the same fieldbus, and unify their communication interfaces.

Besides wired communication systems, wireless communication technologies have also been considered and deployed in industrial environments for monitoring and open-loop control applications. Among others, some wireless solutions based on IEEE 802.15.4 PHY have been proposed, e.g., Zigbee [23], WirelessHART [24], wireless networks for industrial automation - process automation (WIA-PA) [25], and Smartmesh

IP [26]. These standards define their own medium access control (MAC) to achieve different features, such as real-time communications, or high reliability. However, the limitations of the IEEE 802.15.4 PHY in terms of bit rate or reliability makes its use unfeasible for emerging high-performance applications. Some industrial applications use the digital enhanced cordless telephone (DECT) technology, given its deterministic medium access. However, it requires improvements on communication robustness and reliability [27]. Recently, the DECT technology has received an upgrade in the form of DECT-2020 new radio (NR). It defines an advanced radio interface technology to support both URLLC and massive machine type communication (mMTC) [28]. However, it supports short-range communications with limited data rates and marginally meets the minimum reliability requirements [29].

Additionally, low power wide area (LPWA) technologies also gained attention in providing communication for emerging markets such as industrial networks [30]. The two leading LPWA technologies are the narrowband-IoT (NB-IoT) [31] and long range (LoRa) cost [32]. Both LoRa and NB-IoT technologies are energy-efficient though they are focused on low data rate and do not provide the expected performance level required by high-end applications.

Regarding high-performance wireless technologies, there is a clear research trend addressing modifying 802.11 to enable deterministic communications. These solutions typically rely on commercial 802.11 chips combined with a customized software (SW) stack to enable time synchronization and use time division multiple access (TDMA) for channel access. Some of the most relevant solutions in this category include IsoMAC [33], RT-Wi-Fi [34], and WIA-FA [35]. However, these technologies provide latencies in the range of several milliseconds, which cannot compete with the wired fieldbuses capabilities. In recent years, some researchers have proposed the use of system-on-chip field programmable gate array (SoC-FPGA) devices to build tailor-made PHY and MAC layers optimized for specific applications [36]. However, the implementation complexity of these solutions is significant and the cost of SoC-FPGA platforms is typically much higher than using commercial devices.

Besides the communication technology, several application layer protocols for lightweight message exchange are used. The application layer handles data formatting and its presentation. Example of application layer protocols may include constrained application protocol, message queue telemetry transport, extensible message and presence protocol, advanced message queuing protocol, representational state transfer, data distribution service, and Java message service [37]. Although, design, deployment, and implementation of an application layer protocol for industrial use cases is another challenge to consider, this challenge is beyond the scope of this paper.

III. TIME SENSITIVE NETWORKING

In the last decade the institute of electronics and electrical engineers (IEEE), the international electrotechnical commission

(IEC), and the internet engineering task force (IETF) have proposed new standards to enable deterministic communications to the Ethernet standard. The IEEE 802.1 TSN TG evolved from the IEEE 802.1 audio-video bridging (AVB) TG to cover the expanded goals and meet the demands of the industrial control community and the automotive community [38]. Such demands have resulted in mechanisms such as time-aware scheduling to guarantee packet delivery with bounded latency, high reliability, and low latency fluctuations (jitter). The IEEE 802.1 TSN TG has focused efforts not only on defining and proposing new standards but also on extending and improving the already available standards like the IEEE 802.1Q [39] and the IEEE 802.1AS for accurate time synchronization [40]. The TSN standards have been designed to enable modular systems, whereby the implementations for a specific application can be tailored to meet its hardware and software requirements. Another motivation behind the development of TSN was to provide a standardized technology that is open – not affiliated to any organization or company [12]. Hence, allowing interoperability between standard conforming industrial devices from any vendor. This added advantage of interoperability on top of deterministic and high reliability communications has pushed the development on TSN forward. TSN in its essence enables the convergence of IT and OT systems, which were previously kept separate given the difference in their communication, safety, and security requirements. TSN also supports direct convergence with higher-layer protocols (e.g., open platform communications unified architecture (OPC-UA) [41], [42]), since TSN is implemented self-contained on the data link layer (layer 2 of the OSI model).

To reach the targeted capabilities, the TSN technology is based on five key features: accurate time synchronization, traffic shaping and scheduling, ultra-reliability, resource management, and flexible network configuration models. Such features are defined across different standard or amendments, as summarized in Table 1.

Since the TSN family of standards gather many mechanisms which result in different capabilities, different standardization bodies are working together to define different TSN profiles for specific application areas. The goal is to ease the configuration of TSN network and provide a reference for the vendors by defining mandatory and optional TSN features. TSN profile standards include IEEE 802.1BA-2011 for AVB networks [43], IEEE 802.1CM-2018 TSN for fronthaul [44], and its amendment on synchronization enhancements IEEE 802.1CMde [45]. The ongoing standardization efforts for TSN profiles includes IEC/IEEE 60802 TSN profile for industrial automation, P802.1DG TSN profile for automotive in-vehicle Ethernet communications, P802.1DF TSN for aerospace onboard Ethernet communications, and P802.1DF TSN profile for service provider networks [4].

In the following subsections we overview the most relevant TSN standards classified according to the feature they provide.

TABLE 1. The TSN Features and Respective Completed Standards With Publication Year. The Ongoing TSN Standards are Not Mentioned

TSN components	Standards
Time synchronization	Timing & synchronization (802.1AS-2020)
Traffic shaping & scheduling	Scheduled traffic (802.1Qbv-2015) Frame preemption (802.1Qbu-2016) Credit based shaper (802.1Qav-2009) Asynchronous traffic shaping (802.1Qcr-2020) Cyclic queuing & forwarding (802.1Qch-2017)
Reliability	Frame replication & elimination (802.1CB-2017) Path control & reservation (802.1Qca-2015) Per-stream filtering & policing (802.1Qci-2017)
Resource management	Stream reservation protocol (802.1Qat-2010) TSN configuration (802.1Qcc-2018) YANG model for Bridging (802.1Qcp-2018) YANG model for CFM (802.1Qcx-2020)
Network Configuration	No particular standard has been defined, but 802.1Qcc-2018 supports the flexible network configuration

A. TIME SYNCHRONIZATION

The IEEE 802.1AS [40], [46] standard defines the procedure to achieve a common sense of time shared among all networked devices including switches, bridges, and end stations in the TSN network. This shared sense of time is a fundamental premise not only for the correct functioning of other TSN mechanisms (e.g., TAS) but also for coordinated functioning of different end stations.

The IEEE 802.1AS provides a generic precision time protocol (gPTP) profile for accurate time synchronization, using IEEE 1588v2 precision time protocol (PTP) [47]. PTP uses the master-slave paradigm, where a device with the most accurate time source is selected as the grandmaster (GM). The GM's clock is then used to provide time reference to other devices in the network. PTP sends periodic sync and followup messages from the clock master (CM) egress ports to the clock slave (CS) ingress ports. This mechanism synchronizes the local clocks of the networked devices with each other. In order to ensure more stringent synchronization, the standard also introduces redundancy in GM devices and synchronization hierarchies by supporting multiple gPTP domains. The redundant domain is used when the performance of the first domain degrades.

B. TRAFFIC SHAPING AND SCHEDULING

A TSN stream is defined as a data flow with a given QoS requirements between two end stations. Based on the QoS, Each flow is uniquely identified by using the priority code point (PCP) field (traffic class) and the VLAN ID (VID) within the IEEE 802.1Q tag in the Ethernet header. IEEE 802.1Q [39] defines up to eight traffic classes per Ethernet port, where a dedicated queue is associated with each traffic class. Each queue can then be controlled by different TSN mechanisms that results in different QoS provisioning, such as TAS (IEEE 802.1Qbv amendment) for time-triggered communications, or

credit based shaper (CBS) (IEEE 802.1Qav amendment) for priority-based transmissions.

IEEE 802.1Qav [48] ensures fair distribution of frames to queues by shaping the network traffic so as to control the latency and bandwidth of the reserved streams. It defines a transmission credit to each queue, whereby a queue is eligible for transmission if the transmission credit is non-negative and no frames from any other queue are being transmitted. The credit is increased for a non-empty queue if no frame is transmitted, and decreased when a frame is transmitted.

IEEE 802.1Qbv ensures bounded latency for hard real-time traffic by defining a schedule for the whole network [5]. It uses the TDMA paradigm to transmit messages in allotted time windows (i.e., time aware traffic window) and a dedicated queue is associated with each traffic class. A transmission gate is defined for each queue. The transmission schedule is defined in the form of a gate control list (GCL), which controls the state of the gates (open or closed) and defines the duration for which a gate is open (time-aware traffic window). In each cycle the GCL is executed once and is repeated for subsequent cycles. The transmission of frames is controlled by both the GCL and the transmission selection algorithm. The transmission selection algorithm ensures that low-priority traffic does not interfere with high-priority traffic and allows transmission of high-priority traffic first, if multiple gates are concurrently open.

The unpredictable pattern of best-effort traffic gives rise to the priority inversion problem, i.e., the on-going low-priority traffic prevents transmission of high-priority traffic. To address this, IEEE 802.1Qbu frame preemption specifies suspension or holding of the ongoing low-priority transmission if a high priority queue has a frame to send [6]. The IEEE 802.1Qbu is connected to IEEE 802.3br which defines two MAC service interfaces namely the express interface and the preemptable interface. Frames mapped to preemptable interface may be preempted by express frames while frames mapped to express interface cannot be preempted. Moreover, preemptable frames cannot be preempted by other preemptable frames regardless of their priority.

Additionally, the IEEE 802.1Qch standard specifies cyclic queuing and forwarding (CQF) to assign egress queues to frames based on their arrival time, which is essential for receiving all transmitted frames during the expected cycle [49]. CQF along with TAS also ensures deterministic delay for all frames.

In contrast to the standards mentioned in this section, the IEEE 802.1Qcr (asynchronous traffic shaper (ATS)) operates without the need for any synchronization between networked devices [50]. ATS schedules traffic based on its urgency. To implement this, it reshapes TSN traffic streams per hop to smooth out the traffic patterns, implements per-flow queues, and prioritizes urgent traffic over relaxed traffic. These mechanisms help achieve high bandwidth efficiency and low delay, but also imply a high computation process on each frame.

C. ULTRA RELIABILITY

Reliability is an important requirement for many TSN applications. Although, cyclic redundancy check can detect errors and notify the sender for re-transmission, it causes delays that are not tolerable by time critical flows. Therefore, enabling fault tolerance in the network increases reliability and supports deterministic communication. TSN comprises some standards to enhance the reliability of the network and add fault tolerance. Those standards are described as follows.

The IEEE 802.1CB standard for frame replication and elimination provides proactive redundancy in order to decrease packet loss probability. It identifies packets belonging to a stream, replicates them for transmission over possibly disjoint paths, and removes the duplicate frames at the destination [51]. The IEEE 802.1CB is similar in properties and compatible to the well-known parallel redundancy protocol (PRP) [52] and high-availability seamless redundancy (HSR) [53] protocols, thus providing inherent interoperability.

The IEEE 802.1CB does not define disjoint paths itself, instead relies on the IEEE 802.1Qca [54] standard for path control and reservation. IEEE 802.1Qca builds on the intermediate system-to-intermediate system (IS-IS) [55] to create multiple paths through the network. It offers explicit path control, redundancy for data flows between stations (STAs), stream & bandwidth reservation, and distribution of control parameters for flow control messages and flow synchronization [54].

The IEEE 802.1Qci standard employs the stream identification of IEEE 802.1CB to implement per-stream filtering and policing based on rule matching. It allows for QoS provisioning by detecting which streams violate the defined behavior and takes mitigating actions [56].

D. RESOURCE MANAGEMENT

Resource management is a key aspect to achieve deterministic and reliable communication. It defines mechanisms and services to manage resources including bandwidth, scheduling patterns, and communication paths. The following standards in TSN contribute towards resource management.

The IEEE 802.1Qat standard for stream reservation protocol (SRP) provides protocols to ensure resource availability along the entire path between the source and destination. It achieves this by allowing resource allocation and reservation at each switch, thus ensuring end-to-end latency and bandwidth guarantees [57].

Apart from the distributed management by 802.1Qat, TSN also specifies the IEEE 802.1Qcc standard to globally manage the network. It provides an enhancement to the existing SRP by defining new architectures and specifying a software-defined interface between the user and the network called user/network interface (UNI) [58]. Additionally, UNI also defines a fully distributed and a hybrid architecture model. For the fully distributed model, the users send their requirements directly to the network through the UNI. The SRP

includes an application called multiple stream reservation protocol (MSRP). The MSRP describes the specific attributes of a station to transmit or receive streams if they are either Talker or Listeners. Since MSRP has some limitations, the MSRP gen 2, also named MSRP++ has been created [59]. MSRP++ overcomes those drawbacks by splitting the configuration information into static and dynamic and by adding new fields about the stream properties.

The TSN TG also defines IEEE 802.1Qcp to enable communication of TSN configuration data between bridges using YANG models [60]. YANG is a data modeling language used for data configuration, remote procedure calls, and notifications for network management tools. Examples of YANG based protocols include RESTCONF and NETCONF. Apart from this IEEE 802.1Qcx represents YANG models defined for connectivity fault management (CFM) for relays and bridges [61].

E. NETWORK CONFIGURATION

A TSN network consists of a set of interconnected full-duplex (TSN) Ethernet switches (forming a multi-hop topology); and a set of (computational) TSN nodes (each connected to one or more TSN switches). The configuration of such network is a complex task and it depends on the TSN network architecture models. Three models currently exist: fully centralized, hybrid, and fully distributed, as defined in the 802.1Qcc amendment [58]. Therefore, the network configuration relates to the TSN configuration models and the network topology where it is applied to. For the fully centralized model, a centralized user configuration (CUC) entity is defined along with the centralized network configuration (CNC). The UNI is defined between the CNC and the CUC to exchange the information related to the user specific communication requirements. For the hybrid model, the user requirements are communicated directly to the network, but the configuration information is sent by the CNC. There is no CUC entity in the hybrid model. 802.1Qcc is complemented by 802.1Qca [54] which defines the stream reservation capabilities, including path control, bandwidth, and stream reservation of the TSN flows. Finally, 802.1Qci [56], which is also used to enhance the network's reliability, enhances the current existing protection and filtering mechanisms to make sure that the nodes stays with the expected schedulers and to avoid interference from external network devices. Moreover, the IEEE P802.1Qdj as an ongoing TSN standard specifies procedures and interfaces to enhance the three network configuration models.

IV. WIRELESS TSN

The ongoing work on wired TSN promises deterministic latency along with high reliability. However, wired networks lack flexibility given the high maintenance and installation costs of wired connections and minimal support for mobility. With the advancement of industrial technologies, a new vision (often termed as industrial internet of things (IIoT)) is emerging where all industrial assets are interconnected with each other including the cloud, the information systems, and

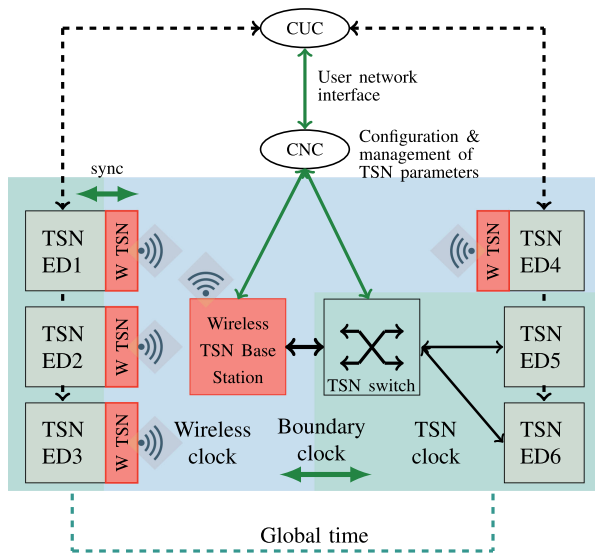


FIGURE 1. Simplified wireless TSN network fully centralized.

the business processes. Wireless technologies are the key enablers to realize this vision for factories of the future. The benefits of adding wireless includes inherent support for mobility, possibility to monitor harsh environments, flexible and modular production systems that support mass customization, and ability to self-configure after failures [7], [8], [62], [63].

A. ENABLING WIRELESS TSN INTEGRATION

Deterministic latency and high reliability performance are the prerequisites of any communication technology in the IIoT. The deterministic latency and high-reliability are mostly fulfilled by wired fieldbuses, though wireless protocols struggle to fulfill them. The main challenge comes from the error-prone nature of the wireless channel and the fact that the wireless media is shared by all the devices.

With the advancement of industrial technologies, a new vision (often termed as IIoT) is emerging where all industrial assets are interconnected with each other including the cloud, the information systems, and the business processes. Industrial assets include sensors, actuators, robots, machines, and control systems [64], [65] (see Section VII-A for a brief overview of IIoT applications). We focus on the challenges at MAC and PHY layers, given that industrial communication systems are typically focused on these layers along with the application layer [66].

The components and functions of a fully centralized TSN network must include TSN enabled switches, end devices (EDs), TSN clock, CUC, and CNC. To support IIoT use cases for additional advantages of mobility and flexibility, extending TSN capabilities over a wireless network is essential. Therefore, a step forward is to enable a hybrid wired-wireless TSN network. Considering this scenario, some components of a TSN network such as TSN EDs are to be replaced with wireless TSN EDs and TSN switch replaced with a wireless TSN base station (BS), as presented in Fig. 1. The wireless TSN

BS comprises both the BS and core network in case of 5G as wireless technology while in case of Wi-Fi is a TSN capable access point (AP). However, a simple substitution of wired to wireless components is not enough if there is no coordination between the wired and wireless domains and if the wireless domain does not support TSN capabilities. In general three different layers or functional groups can be identified that may be addressed for transition to a hybrid wired-wireless TSN network. These include PHY-layer, link layer with time and communication synchronization, and network configuration regarding services, QoS, latency, and similar.

On the PHY-layer, TSN capabilities can be extended over a wireless network through interfacing between the wired EDs and the wireless EDs. This can be established either through a wireless bridge such as the 5GS (5G core and BS), providing wireless interfaces such as connecting APs, or through a wired connection such as Ethernet to a WiFi BS. Here, we use the general term BS to address the media conversion from the wired to the wireless domain and vice versa. Two types of BSs can be identified, namely, a simple BS and a complex BS. Simple BSs are placed at the ED ends and allows allows interfacing between wired and wireless TSN EDs, e.g., ED1-ED4 to a wireless channel, as presented in Fig. 1. A complex BS translates the serial TSN Ethernet communication to the wireless domain. This BS has to act as a wireless bridge with the possibility to analyze or compile Ethernet data and respectively send/receive the data in defined time slots using the corresponding wireless channels. The mapping between Ethernet data and wireless channel time slots (irrespective of the technology) is to be configured as part of the link layer functions.

The link layer allows to link packets of the Ethernet stream according to their identification to specific time slots in wireless channels. However, the forwarding of the traffic with low, deterministic latency, requires tight time synchronization between the domains. In Fig. 1, the wireless domain is depicted in blue and wired in green where time synchronization of the two domains can be performed using a boundary clock.

It is expected that the functionality for CNC and CUC remains unchanged except for the parameters that are specifically required for configuration of the wireless links and the ones that are beneficial in establishing communication links with desired performance. Once the PHY layer and link layer mapping between the wired and wireless domains is established, ensuring QoS along with enabling resilience, security, and efficiency would be required. In the following, we present the main challenges to be tackled for enabling a wireless TSN system.

B. CHALLENGES TO SUPPORT WIRELESS TSN

To support wireless TSN, several research challenges can be foreseen, which are mostly related to the wireless medium. First, the wireless medium, specifically, under the unlicensed frequencies is contention based and poses the challenge of channel access due to the listen before talk procedure. However, guaranteed access to the wireless channel is essential to

enable deterministic communications. The demanding question is to design mechanisms that enable guaranteed channel access in unlicensed frequency spectrum. A related issue of the wireless medium is the inherent interference from other channels. The interference and signal distortion issues pose challenges in achieving ultra-reliability. In the first place, interference from nearby wireless devices that communicates with the same protocol may be avoided by a good protocol design (e.g., using coordination between devices). Interference from wireless devices using other protocols are more complex and mechanisms to minimize interference are required. For instance, in the case of a factory, the mechanisms to minimize interference could be shielding the walls of the factory or implementing a strict access control of devices so no one transmits in the operation bands of the industrial devices. Also, the Wireless TSN can implement functionalities to detect interference and move to other bands, as is the case in cognitive radio approaches.

Apart from the inherent problems of interference in unlicensed bands, common wireless propagation phenomena also poses a significant hurdle in the achievable reliability and latency. Therefore, the propagation phenomena in the target environments must be taken into account in the design and implementation of the protocol, and in the deployment of the wireless network infrastructure.

To enable TSN integration with wireless technologies, several challenges are to be addressed. These include enabling time synchronization, time-aware scheduling, providing the ultra-reliability targets required by the applications [12], [13], [17], [67]. Moreover, exposing the wireless QoS capabilities from the wireless TSN to the wired TSN domain and its integration with the TSN configuration and management is another challenge. A crucial feature of TSN is to support vendor independent network configuration. These can be configured through protocols such as NETCONF to ensure that configuration of the TSN features is performed according to standard methods. However, in case of any changes such as installation of additional sensors, a manual re-configuration is required. Therefore, the challenge is the design of re-configurable system that allows changes in the network yet enabling smooth network functioning. A re-configurable system, however, also requires defining interfacing and interaction between the sensors, the wired TSN domain and the wireless TSN domain. The configuration setting is also to be reflected at the wireless system in a dynamic manner for automatic deployment and configuration of an end-to-end hybrid wired-wireless TSN system. In the following sections, we detail how some of these challenges can be addressed by IEEE 802.11 and 5G.

V. ENHANCEMENTS TO SUPPORT TSN IN IEEE 802.11AX/BE - WI-FI 6/7

During the past few years, the IEEE 802.11 standardization groups have been working on the 802.11ax standard (Wi-Fi 6) [9]. Efforts have not only been focused on providing higher throughput, but also on enhancing some other aspects such as latency, reliability, and flexibility. The most relevant

features are the modification of the orthogonal frequency division multiplexing (OFDM) numerology, the introduction of orthogonal frequency division multiple access (OFDMA), and the definition of the TF and target wake time (TWT) mechanisms. The new 802.11ax mechanisms enable significant latency reduction and enhanced efficiency, though IEEE 802.11ax does not still reach the performance figures in some important TSN aspects, such as time-aware scheduling and ultra-reliability [13]. The new 802.11be standard (Wi-Fi 7) is under definition and its final specification will be released in two to three years [68]. An expected novelty for 802.11be is the introduction of the low-latency operation mode [69]. The low-latency mode may provide the QoS required by TSN-capable systems and so enabling wireless TSN [16].

Through this section, we first comprehensively analyze all the current existing mechanisms in 802.11ax and the new mechanisms that are yet to come in 802.11be that can be used to enable wireless TSN. Then, taking into account that the integration of wired TSN and 802.11 is yet to be defined, we propose a possible 802.11-based TSN architecture that includes the configuration of the 802.11 mechanisms to enable wireless TSN and the integration of such mechanisms with wired TSN.

A. TIME SYNCHRONIZATION

The problem of time synchronization using 802.11 has been widely studied in the literature. Two elements are key to successfully enable time synchronization for wireless TSN: the technologies/protocols involved in the actual wireless time synchronization and its coexistence with other wireless TSN mechanisms (e.g., traffic shaping and scheduling).

1) WIRELESS TIME SYNCHRONIZATION

The attainable wireless time synchronization depends mainly on three variables: the messaging scheme of the synchronization protocol, the timestamping technique / available clock, either SW or hardware (HW) timestamps/clock, and the specific wireless conditions. The largest error source is the timestamping and clock, whereas the messaging scheme and the wireless channel phenomena are typically a source of small time synchronization error.

Two main families of time synchronization messaging schemes exist: one way messaging based on the broadcasting of time frames (e.g., beacon frames) or two-way, when the slaves acknowledge the time frames. The latter is able to compensate the channel delay thanks to the two-way exchange, whereas the former cannot.

In the particular case of 802.11, the timing measurements (TM) and fine timing measurements (FTM) define two nearly identical messaging schemes to enable precise time synchronization. Fig. 2 represents the FTM frame exchange, which is described as follows. First, an STA sends a FTM request message to start the synchronization procedure. The AP answers the request with a FTM frame. The STA receives the FTM and answers an 802.11 acknowledgment (ACK). During the frame

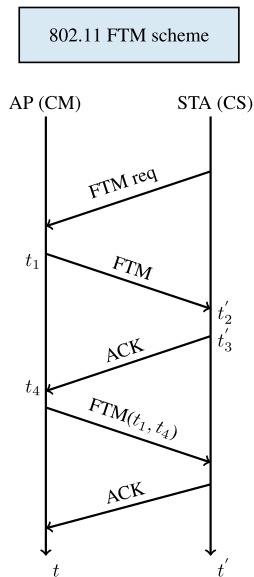


FIGURE 2. Synchronization schemes considered in 802.11ax for wireless TSN operation.

exchange, a set of timestamps is collected, which represent the frames ingress and egress times. The frame exchange is repeated several times (depending on the FTM configuration) to take more timestamps. Finally, the timestamps are used by the STA to compute the synchronization error and frequency drift and compensate its internal clock time. The frame exchange is periodically performed to continuously correct small variations in the clock drift.

The timestamping technique is the strongest limiting factor in the synchronization performance [70]. The timestamping techniques are classified as SW or HW. SW-based timestamps are affected by the jitter of the communication stack and so they provide low performance, in the 1 μ s precision range [71]. In comparison, HW timestamps reaches synchronization levels in the order of 10-40 ns for 20 MHz BW and over different channel conditions, as highlighted in several research works [36], [72]–[75]. Some of these works highlight that the wireless propagation phenomena is one strong limitation in the achievable synchronization level. Even so, the achieved performance for the worst-case channel conditions, is enough to reach the performance targets of wireless TSN.

2) COEXISTENCE OF TIME SYNCHRONIZATION AND OTHER TRAFFIC STREAMS

Apart from the performance itself, the coexistence of the time synchronization with other traffic streams is also a relevant matter of study. It is extremely important that the timing messages do not impact the scheduling of the TSN flows. At the same time, the traffic from the TSN flows should not block the timing messages. The natural option would be using the 802.11 FTM. However, an unauthenticated and non-synchronized STA may cause interference to the wireless TSN traffic flows when it attempts to send FTM request frame

to synchronize its timer to the network base time. A hybrid synchronization using the 802.11 beacon frames and the FTM scheme could be a suitable solution. In the first step, in an unsynchronized state, the STAs could perform a coarse-grain synchronization exploiting 802.11 beacons. In the second step, after the coarse synchronization, it could switch to the 802.11 FTM.

B. TRAFFIC SHAPING AND SCHEDULING

802.11 default medium access is contention-based where every node of the network can gain access to the wireless medium at any time. Such medium access is not deterministic and cannot virtually provide any level of QoS. Nonetheless, 802.11 includes other operation modes that enable traffic classification and contention-free based medium access. Regarding traffic classification, 802.11 includes a medium access scheme named enhanced distributed channel access (EDCA) [9], which includes four traffic classification profiles: video, voice, background tasks, and best-effort. Even though these traffic profiles are not aligned with the TSN needs, they may be redefined or extended for the TSN needs, given that, real applications may roughly require three traffic priorities (hard real-time, soft real-time, and best-effort). Besides, other mechanisms such as internet protocol (IP) classification [13] are also being considered to perform traffic classification in 802.11be. Nonetheless, TSN sub-standards are defined just for the second layer of the communication stack and thus the use of IP may constraint the design of custom stacks for specific applications.

Concerning the time-aware operation and QoS, 802.11ax supports several methods to prevent undesired transmissions from the STAs connected to a wireless TSN-capable AP and to ensure almost deterministic transmissions. Nonetheless, it is worth mentioning that since 802.11 is based on the listen-before-talk procedure, the transmissions will always present a jitter (even for free channel conditions). The jitter should be below 100 μ s for the common 802.11 random access procedure when no interference is present. The next medium access mechanisms currently being considered to enable time-aware operation in 802.11ax include EDCA, hybrid coordination function controlled channel access (HCCA), TWT, and TF-based [13]. EDCA and HCCA were already included in the older versions of 802.11 standard, whereas TWT was first introduced in 802.11ax. In addition to the scheduled TF, an enhancement to the 802.11ax TF is currently under consideration by the 802.11be task group (TGbe) group [13]. These medium access mechanisms are based on a similar principle that the AP prevents the transmissions of the STAs and establishes priorities based on the real-time traffic needs of the STAs. The most promising medium access may be the scheduled TF, which enables a very tight scheduling of the ongoing transmissions.

In the first place, EDCA, and HCCA have been widely studied in the literature over older versions of the standard. Most

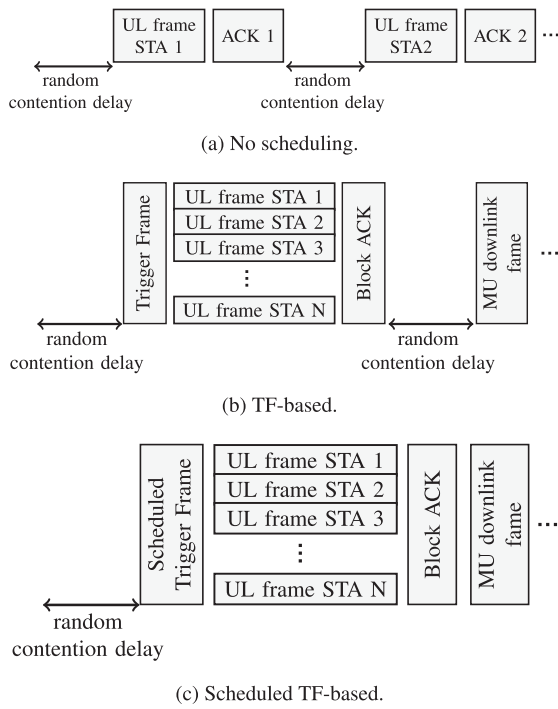


FIGURE 3. Comparison of 802.11 scheduling modes.

researchers concluded that they are not suitable to handle real-time traffic [76]. In the second place, the TWT, originally designed as a power-saving mechanism, is also being considered to enable time-aware operation. The TWT mechanisms enable the activation/deactivation of STAs to reduce their consumed power [18]. This feature can also be used to wake up the STAs at different time instants so that the STAs will compete for the medium with fewer STAs.

Finally, the TF-based scheduling uses the TF to deterministically schedule 802.11 frames [13]. First, the AP must gain access to the wireless medium using the 802.11 legacy random access mechanisms. Once the AP has gained the medium access, the AP can transmit a TF that is used by the 802.11 nodes to access the medium without an additional contention. As shown in Fig. 3(a) and (b), the TF significantly reduces the number of random contention waiting periods between frame transmissions, thus effectively enhancing the network efficiency and latency. TF enhancements are under consideration for 802.11be to enable time-aware scheduling. Basically, the idea is to replace the TF with an scheduled TF that enables the transmission of several frames in a row, as shown in Fig. 3(c). The scheduled TF may significantly reduce the communication jitter and latency since the AP must only gain access to the channel once, as opposed to using the legacy TF.

C. ULTRA RELIABILITY

Achieving ultra-reliability is one of the most challenging features in wireless communications over unlicensed bands because 1) the wireless channel heavily distort the wireless signals and 2) unlicensed wireless bands are shared among

different wireless technologies and users and so there is a high probability of interference. The ultra-reliability challenge is typically tackled by combining two complementary perspectives: intra-frame redundancy and inter-frame redundancy.

1) INTRA-FRAME REDUNDANCY

Intra-frame redundancy introduces redundant bits within a frame, reducing the data rate though effectively enhancing the probability of successful reception of a frame. The ratio between the number of bits (redundant + data) and the number of data bits transmitted in a frame dictates the protection level of a frame against the wireless impairments. At this moment, the most reliable modulation and coding scheme (MCS) in 802.11ax is BPSK with redundancy of $1/2$ (MCS = 0). In addition, 802.11ax has also introduced the dual carrier modulation (DCM) to enable the transmission of redundant data over different subcarriers to enhance the reliability under high-multipath channels. Unfortunately, to the best of our knowledge, there is no work available in the literature that analyzes the reliability of 802.11ax using DCM.

Some researchers have comprehensively studied the reliability of intra-frame redundancy using solutions based on the 802.11 PHY for industrial environments [36], [77]. These studies show that $PER < 10^{-6}$ using single-frame transmissions are feasible in industrial environments using robust modulation schemes and decoder implementations. However, the presented results are for environments with no external interference.

2) INTER-FRAME REDUNDANCY

Intra-frame redundancy is absolutely necessary to ensure an appropriate reliability of the wireless TSN system. However, it does not provide enough protection under situations with interference or in the case of incorrect encoding/decoding due to a fault in the transceiver electronics. Therefore, the use of inter-frame redundancy is almost compulsory to reach the reliability targets required by industrial applications. The inter-frame redundancy is typically classified into active redundancy and passive redundancy. In active redundancy, the transmitter detects through an ACK frame if the data frame has been lost and acts in consequence (e.g., performing retransmissions). In the passive redundancy, different copies of the same frame are transmitted through different resources/paths (either time, frequency, space, or different communication technologies) without waiting for the ACK from the receiver.

Active redundancy is spectrum-efficient as it only transmits repetitions of a frame if the initial frame is lost. In addition, the repeated frame can be shortened using hybrid automatic repeat request (HARQ) techniques, minimizing the amount of radio resources consumed in the retransmission. However, it clearly impacts the achievable latency because the retransmissions require signaling from the receiver to know if the frame was correctly delivered or not latency [78].

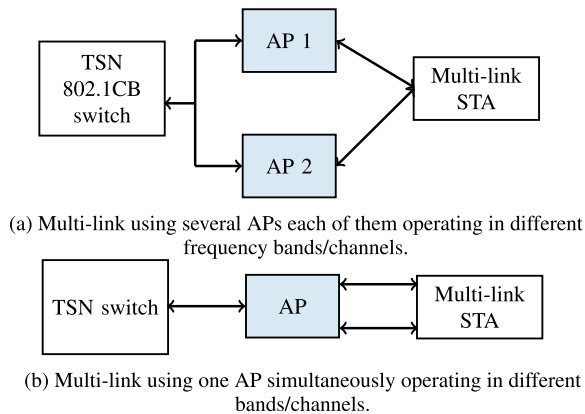


FIGURE 4. Multi-link Redundancy techniques proposed for 802.11be.

On the contrary, passive redundancy has lower efficiency, since multiple repetitions of the same frame are transmitted through different channels. Nonetheless, it is a very compelling method to achieve ultra-reliability in wireless TSN without sacrificing latency and without requiring complex retransmission schemes. Passive redundancy is currently not supported in 802.11ax, however, is being considered for the 802.11be standard using multi-link operation [79].

The multi-link operation allows an STA to simultaneously maintain multiple 802.11 links across the 2.4, 5, and 6 GHz bands. The multiple links can be used to send redundant frames as in 802.1CB standard (see Fig. 4). Multi-link is expected to significantly enhance the network reliability. Multi-link will be implemented in a new entity named multi-link management, built on top of the 802.11 MAC layer. The multi-link management and TSN entities will configure each link based on the traffic QoS requirements and the wireless link capabilities. Finally, it is worth noting that multi-link may also enhance the average and worst-case latency in unlicensed bands [80]. Basically, each 802.11 link will operate under uncorrelated interference and propagation conditions. Therefore, the probability of transmitting one of the redundant copies of the frame before the specified time is increased.

D. RESOURCE MANAGEMENT

The management and allocation of resources is always a critical issue in industrial wireless networks, since resources such as spectrum and transmit power are limited. Motivated by this, there is a large body of work on managing such resources and improving the overall performance of the system. The OFDMA modulation provided by 802.11ax allows efficient transmissions of small frames to a group of users simultaneously. The whole bandwidth is divided into multiple subsets of subcarriers, each subset is known as a resource unit. To efficiently utilize the available bandwidth, one needs to optimally solve the scheduling and resource allocation problem. Several proposals can be found in the literature with different approaches to optimize the user and spectral resources [81], [82].

In addition to the previous 802.11 standard, 802.11be proposes the introduction of a new feature related to the management of a complete network that comprises several APs, named as multi-AP resource coordination. Among others, a purpose of this feature is to manage the mutual interference and contention levels between different AP networks and optimize the deployment of the network. To do so, it assumes a centralized global management of the entire network, named the Master AP, where all the wireless devices belong to the same logical network.

Finally, apart from the intrinsic challenges of the resource management of the wireless protocol, the interference from other networks in unlicensed bands is a significant extra challenge for the resource management. This challenge is even more significant when the networks has to fulfill stringent QoS capabilities. In that sense, adequate network planning of the wireless technologies is extremely important to minimize the wireless interference [83]. Recently, the available spectrum for 802.11ax has been extended in some countries by adding 1 GHz extra bandwidth [84]. This band is currently free of other wireless systems, therefore, it may be an interesting opportunity to avoid interference and ensure appropriate network planning. 802.11ax devices that operates in 6 GHz band have been commercially labeled as Wi-Fi 6E.

E. NETWORK CONFIGURATION

Traditionally, 802.11 is based on a star network topology, centralized around an AP, which behaves as a gateway for the exchanged frames between the devices. There is also a standard extension for supporting mesh network topology [85]. Its main objective is to enlarge the network coverage using different hops between the AP and STA. Additionally, 802.11be will provide an evolved network configuration based on simultaneous AP links for each STA, which can be considered for improving the throughput or even increasing the logical link reliability. This feature will require perfect coordination between the APs and consequently another wired network configuration around the APs.

Besides the standardized solutions, there have been several proposals for improving the coordination within the APs. The authors in [86] consider the mechanisms for the handover process of the STA when it changes from one AP to the next one. From the reliability point of view, the authors from [87] and [88] propose two parallel 802.11 networks following the approach defined as parallel redundancy protocol (PRP) [89]. These solutions involve new network architectures not only in the wireless domain but also in the wired domain that support the correct behavior of the overall network.

F. 802.11-BASED TSN AND THE INTEGRATION WITH WIRED TSN

Even if next 802.11 standard releases supports Wireless TSN, 802.11 as its own is not enough to fulfill all the communication requirements in real deployments and so it has to rely on a wired TSN backbone that interconnects the wireless network with wired devices. Therefore, the last important aspect to

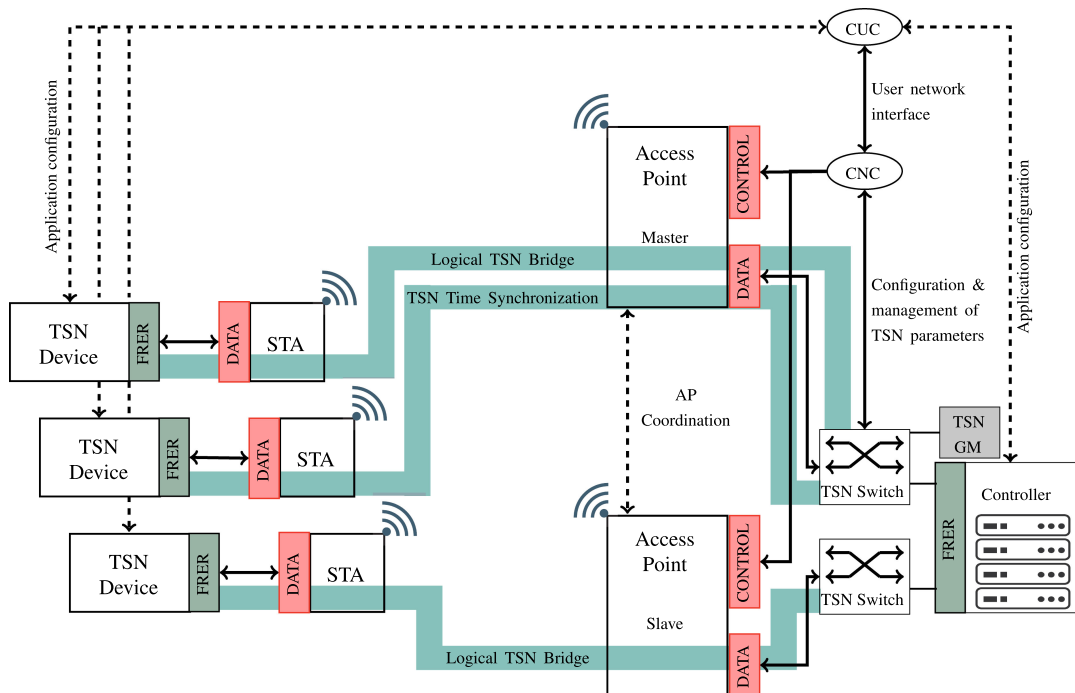


FIGURE 5. 802.11 and TSN hybrid network architecture.

successfully enable wireless TSN is to integrate the wireless TSN technology into wired TSN infrastructure.

Several action points have to be considered to reach a successful integration, as introduced in Section IV. First, 802.11 must support a QoS similar to wired TSN. These aspects have been already reviewed in the previous sections. A second important aspect is the time synchronization translation between the wired and wireless domains. The third challenge is related to translating the frame data and their priority and ID through the domains. Finally, the last challenge is to translate the 802.11 QoS capabilities from the wireless TSN to the CNC language, and from CNC back to the 802.11.

These challenges are serious hurdles to enable the integration of 802.11 with wired TSN. Through this subsection, we have analyzed such challenges and we have developed an 802.11-based TSN network architecture, which is depicted in Fig. 5. The presented architecture includes a common wired TSN segment that includes a set of wired TSN switches, wired TSN devices (e.g., the controller), the TSN CM that provides the global network time, and the CNC and CUC configuration entities that performs the wired and wireless TSN network orchestration. The wired TSN is extended to the wireless domain using 802.11 APs. The APs include the key elements that enable the translation of the TSN traffic and services, namely the data translator and priority mapping, the time synchronization forwarding, and the QoS functions translator (named in Fig. 5 as the control block at the APs). The data translator is an existing element in the Ethernet-802.11 integration, whereas, to the best of our knowledge, the rest of the subsystems, which are required for wireless TSN operation, are not standardized. In the following paragraphs,

we will describe the functioning of each subsystem of the architecture to achieve a high-performance integration.

1) INTEGRATION OF WIRED AND WIRELESS TIME SYNCHRONIZATION

A key point in the integration of wired and wireless TSN is the translation of the time synchronization from the wired to the wireless domain. Basically, the wired TSN must include a CM that propagates the synchronization to every wired TSN capable device of the network including the 802.11 APs (Fig. 5). Then, the APs propagate in turn the timing from the wired to the 802.11 interface. In the case of the integration of TSN and 802.11-based TSN, a boundary clock is probably a better approach because the APs not only require to propagate the synchronization but they also need to be synchronized to the network timing to implement the time-aware scheduling. The integration of the synchronization poses an implementation challenge and a protocol translation challenge. Regarding the implementation, the main challenge is how to share the same clock across the different wired and wireless interfaces. Basically, custom hardware solutions are required to forward the time provided by a hardware clock from one interface to another, as shown in [90]. In addition to this, a second challenge is related to the synchronization protocol. Basically, the wired and wireless domain may use different clock synchronization protocols and therefore may use different time formats (timestamps) as well. As a result, the time has also to be converted to an appropriate format.

A boundary clock-based wired and wireless TSN time synchronization translation has been demonstrated in [90]. The

work shows that the translation can be done with virtually no loss in precision if hardware-based translation is employed.

2) DATA TRANSLATION AND PRIORITY MAPPING

The data translation and priority is the second important challenge of the 802.11 and wired TSN integration. Regarding the data translation, 802.11 is a natural extension to Ethernet and it already includes all the required translation mechanisms to forward a frame from the wired domain to the wireless, such as header parsing and translation, fragmentation if required, etc. However, the priority mapping between 802.11 and Ethernet is not standardized and so vendor-specific solutions are common in commercial APs. This includes, for instance, automatic virtual LAN (VLAN) tagging of the traffic forwarded through Ethernet at the AP.

Two priority mapping models can be envisioned: non-integrated 802.11 priority mapping, and integrated 802.11 priority mapping. In the former one, the mapping is not provided by the 802.11 itself, but by an external hardware or software entity that transmits the frames to the modem according to an external schedule. This model has already been considered in some works that aim to enable TSN over 802.11ac and ax [17]. In the second one, 802.11 is actually priority aware and implements different queues for each traffic stream. The transmission of the frames of each queue is controlled by the specific 802.11 MAC mechanism implemented over the network, such as done in the EDCA mechanism. Such priority classification could be enabled for other schemes, such as the scheduled TF. However, the integrated 802.11 priority mapping requires support from the 802.11 standardization bodies and from the vendors to develop the required specific hardware mechanisms.

In addition to this, it is also necessary to tag the 802.11 frames according to their TSN VLAN tags. To do so, 802.11 includes a QoS field that can be used to know the specific QoS properties of a frame. This field could be used to provide information about which frame belongs to which wired TSN flow.

3) 802.11-TSN QOS MAPPING

Finally, the 802.11-TSN QoS mapping is the last important challenge of the 802.11 and TSN integration. The QoS mapping is refers to two functionalities implemented between the CNC and the 802.11-capable devices. First, the 802.11-TSN capable devices must deliver to the CNC the specific QoS supported by the 802.11 links, including aspects as throughput, minimum latency, worst-case latency, and mean reliability. Since 802.11 QoS does not use the semantics (parameters) specified by TSN, the fundamental problem here is how to correctly translate the information so no details are lost in the process, or how to modify the TSN parameters so they include all the relevant information about the 802.11 link. In addition, the notification of these capabilities and the available mechanisms is of utmost importance. Basically, because the CNC use such information to create the schedule between

the different flows when they cross the wired and wireless domains. Such schedule is designed with the aim of satisfying the QoS targets of the applications. Second, and once that the CNC computes the network scheduler, the 802.11 devices must be able to translate the QoS into the implemented 802.11 mechanisms so that the TSN flows requirements are maintained.

Regarding the former aspect, the 802.11 must be able to measure the specific QoS that it is able to provide and to notify the QoS to the CNC. link layer discovery protocol (LLDP), defined in 802.1AB [91], and extended by 802.1ABcu to support YANG, is commonly used to discover the network and to know the capabilities of each wired TSN switch. However, LLDP does not include fields to define the 802.11 capabilities. As a result, it may be necessary to use proprietary protocols or to modify LLDP so it include wireless-related fields.

The CNC gathers the information about the network topology, including TSN switches, the 802.11 links, and the talkers and listener configurations from the CUC via the UNI. After that, the CNC computes the network's scheduler. Then, the scheduler is sent to both the wired and wireless TSN-capable devices in TSN-like format (see Fig. 5). The control unit of the APs has then to translate the wired TSN configuration into an 802.11 configuration structure. Such translation strongly depends on the available 802.11 mechanisms, capabilities, wireless deployment, and resources available at each AP. For instance, the coordination between APs working on the same or adjacent channels is expected to play a vital role in the scheduling configuration [80]. That is, after the APs receive the TSN scheduler configuration, the master AP gathers all the information about the schedulers and state of the 802.11 links and creates an 802.11 scheduler that satisfies the QoS of each TSN flow. Additionally, both the TSN flows configuration and 802.11 links are subject to changes in the case that new TSN flows are added to the network or if one wireless link is degraded. In that case, the master AP has to recalculate the network schedule on the fly to maintain the QoS of each TSN flow.

G. EXAMPLE 802.11-TSN INTEGRATED SCENARIO

We have used the paper [92] as an use case and configuration example of a Wired TSN - Wireless TSN (802.11-based) network. The specific application is within an industrial use case for smart grids. The requirements of the application are as follows. The application must periodically sample every 500 μ s some sinusoidal waveforms from different points of the grid. Then, the sampled data must be sent with deterministic latency to a controller which has to decide, based on the sampled values, the state of the grid and act in consequence. To enable this use case over Wireless TSN, [92] implements 1) in the wired domain: 802.1AS and 802.1Qbv to provide deterministic communication with known latency from the network configuration. 2) In the wireless domain: a beacon-based synchronization messaging (similar to the Wi-Fi beacon frames), and a TDMA-based scheduling (similar to the scheduled-trigger frame techniques that are expected

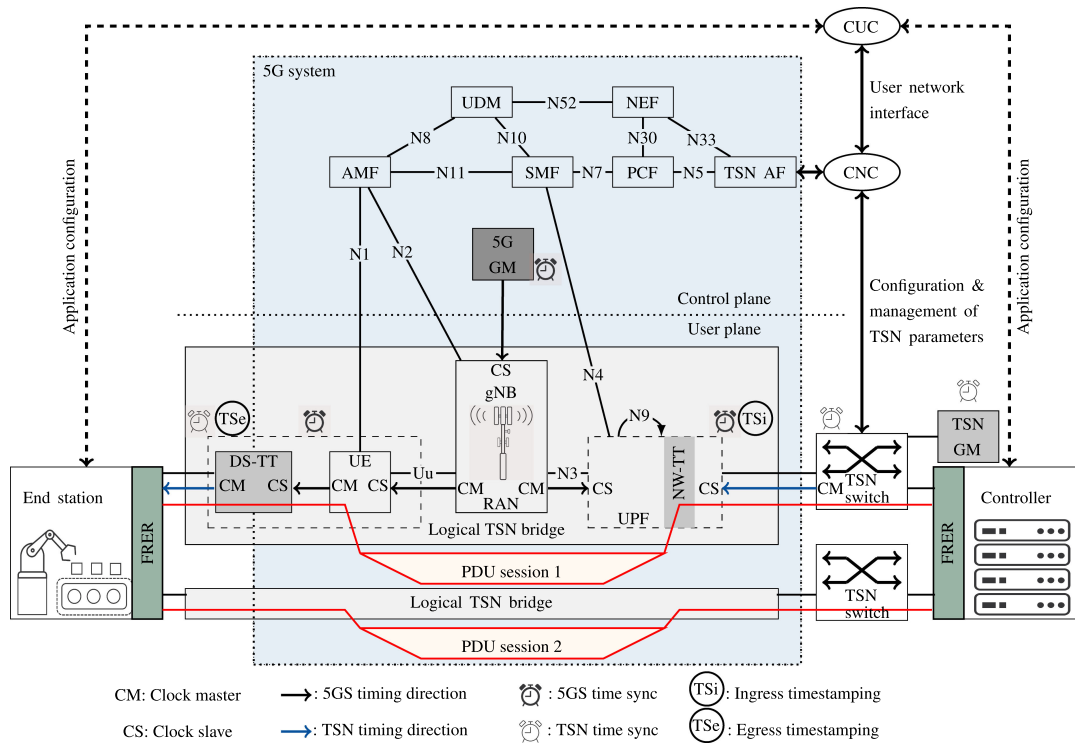


FIGURE 6. 5G system architecture as a logical TSN bridge supporting time synchronization and end-to-end deterministic connectivity.

to be in the next Wi-Fi releases). To ensure deterministic latency, both the 802.1Qbv and the TDMA-based scheduler are synchronized to the network global time. Additionally, the data translation and forwarding between domains is done by hardware means, ensuring minimum and guaranteed latency when the data crosses the domains.

VI. ENHANCEMENTS TO SUPPORT TSN IN 5G

The 3GPP standardization body is continuously working on the evolution of 5G cellular technology. The 3GPP Release-15 focuses on enhanced mobile broadband (eMBB), while in Release-16 and forthcoming Release-17 URLLC gains importance. Besides URLLC, 5G offers network slicing and edge computing that may interest future industrial applications. Moreover, the 5G NR defines a flexible frame structure and mini-slot transmissions based on different numerologies to support low latency communication [93]. The IIoT is a major vertical focus area for Release-16 with features including enhanced support for URLLC, time sensitive communication (TSC), non-public networks (NPN), high accuracy positioning, and NR on unlicensed bands [94].

The 3GPP Release-16 extends support to TSN for applications that demand reliable and deterministic communications. The 5G-TSN integration is applicable to TSN fully centralized configuration model (IEEE 802.1Qcc) whereby the fully distributed and the hybrid (centralized network/distributed user) models are not supported [95]. The 5GS integrates with TSN as a logical bridge where the 5G core and access network procedures remain hidden from the TSN network [67]. The 5GS

architecture as a TSN logical bridge is presented in Fig. 6. The 5G core consists of the control plane and the user plane. The control plane carries signaling and control traffic, whereas the user plane carries user traffic [96], [97]. The signaling and control traffic is referred to as the exchange of information that enables user equipment (UE) access to the network such as authentication, resource allocation, mobility, etc. The user traffic is the actual data that is to be exchanged in corresponding QoS flows. The control plane has functions such as the access and mobility management function (AMF), session management function (SMF), and policy control function (PCF), while the user plane function (UPF) falls under the user plane. The PCF provides policy rules to SMF that include QoS policy and charging control, network slicing, and roaming & mobility management. The SMF performs the session management functions such as allocation of IP addresses to UEs, signaling of QoS and policy information to RAN via the AMF, and selection & control of UPF for traffic routing.

The access network includes the UE and the 5G gNodeB (gNB). 5G TSN logical bridge includes the TSN translator (TT) functionality for both the user plane and the control plane to inter-operate between TSN and 5G system [11], [95]. The TT functionality consists of the DS-TT and the NW-TT. The DS-TT is located at the UE, while NW-TT resides at the UPF. Support for TSN ingress and egress ports is provided via DS-TT and NW-TT. On the control plane, the TT is a TSN AF that interacts with the CNC for control and management. The AF influences traffic routing in the user plane and interacts with the PCF for policy control.

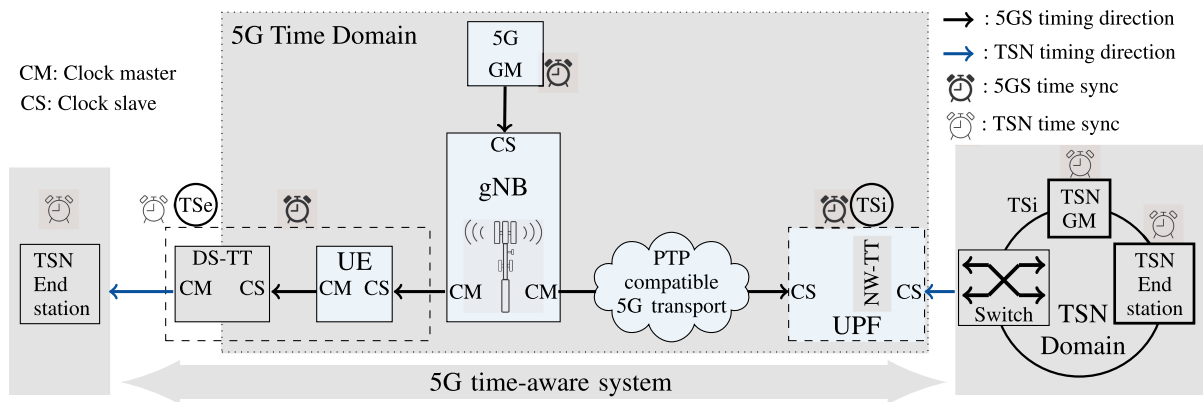


FIGURE 7. 5G time-aware system supporting synchronization through boundary clock or transparent clock solutions [104].

The 5GS virtual bridge comprises the user plane tunnel between the UE and UPF, the NW-TT side port to support connectivity to the TSN domain, and DS-TT side port associated to the protocol data unit (PDU) session that provides connectivity to the TSN network. Only a single PDU session per DS-TT port for a given UPF can be established. However, all PDU sessions that connect to the same TSN network through a particular UPF are grouped under one 5GS virtual bridge. Nevertheless, multiple 5GS virtual bridges can be established using different UPFs that may exist in a 5GS to support redundant traffic transmission [95].

In the forthcoming Release-17 support for fully distributed TSN configuration and UE-UE TSN communication is under consideration [98]. Another viewpoint of adding TSN functionalities in the 5GS instead of TSN-5G integration may also prevail in future [15]. In the following subsections we present the mapping of TSN standards as integration to 5GS for the centralized configuration model.

A. TIME SYNCHRONIZATION

Time synchronization being the fundamental prerequisite for TSN is supported by 5G through establishing Ethernet type PDU sessions between the end stations and TSN switches [95]. Irrespective of the TSN time synchronization, 5G entities such as UE, gNB, UPF, DS-TT, and NW-TT are all synchronized via the 5G internal synchronization process (see Fig. 6). Although 5G and TSN synchronization processes run in parallel, they are independent of each other. The 5G fronthaul synchronization between the radio resource head (RRH) (gNB) and the baseband unit (BBU) is serviced via enhanced common public radio interface (eCPRI) [99], while timing information to the UE is distributed by the gNB through multicast or unicast messages [100]. gNB as a CM, synchronizes the UE and the UPF. The synchronization accuracy requirement of less than $1\text{ }\mu\text{s}$ between the gNB and the UE is specified [101]. Specifically, gNB-UE synchronization accuracy between 470 ns and 540 ns is achievable with 15 kHz sub-carrier spacing (SCS). The synchronization accuracy can further be improved with wider SCS [102]. In addition, some

evaluations suggest that the synchronization accuracy requirement can be achieved without applying UE propagation delay compensation if the inter-site distance (ISD) is smaller than 200 m with dense cell, because the delay is in the range of tens of nanoseconds. Additional details on 5G synchronization process can be found in [103] and are out of the scope of this paper. To support TSN time synchronization with 5GS, boundary clock and transparent clock solutions are considered [15], [105]. The transparent clock solution has been standardized in 5G Release-16 [67], [95]. In addition to the transparent clock solution, Release-17 supports boundary clock solution as presented in Fig. 7. In the boundary clock solution, the TSN GM clock connects directly to the gNB via UPF which in turn distributes timing information to the UEs. The UEs then synchronize their connected TSN devices. This method ensures that every element of the network is synchronized to the same common base time. In this solution, the 5G internal clock can also be used as the time source for PTP GM. Although, it is implementation specific but exchange of PTP messages may not be necessary in this case, as opposed to the transparent clock solution. The transparent clock solution allows the exchange of gPTP messages via TT for synchronization. In this solution the TTs support gPTP and timestamping. As soon as a gPTP message enters the 5GS via the UPF from the TSN domain, an ingress timestamp (TSi) is generated by the NW-TT based on the 5G internal clock and is embedded in the gPTP message. The gPTP message exchange is performed via the established PDU sessions between the UEs and the TSN switch for synchronization of the network endpoints. A gPTP message received by a UE is forwarded to the DS-TT that creates an egress timestamp (TSe). The TSe is also based on the 5GS internal synchronized clock which is provided to the UE by the gNB. The duration that the gPTP packet spends within the 5GS is calculated as the difference between TSi and TSe and is called the residence time. The residence time is added to the correction field (CF) of the TSN synchronization packets. The DS-TT then forwards the modified gPTP packet to the connected TSN devices. The gPTP message payload can be a synchronization message for a one-step operation or a follow up message for a two-step operation. Irrespective of

the gPTP message payload, all gPTP message transmissions within the 5GS are performed in compliance with the QoS requirements of IEEE 802.1AS. However, the timing error is dependent upon the number of hops between the TSN GM clock and the gNB that get accumulated with the higher number of hops [15].

The 5G-TSN integration supports multiple TSN working time domains where each TSN domain can send its gPTP messages. A gPTP message carries the multicast Ethernet destination MAC address and a specific domain number. The domain number identifies the referred TSN time domain. A gPTP message received by the DS-TT for one or more TSN working domain is forwarded to the corresponding TSN device that can select or discard the message depending on the interested domain number.

Release-16 specifies downlink (DL) TSN time synchronization where the TSN GM clock is located at the NW-TT end. On the contrary, the uplink (UL) TSN time synchronization refers to when the TSN GM is attached to the device at the DS-TT side. The UL TSN time synchronization and the UE-to-UE time synchronization are under development in 3GPP Release-17 [98]. The solution proposed is similar to that of the DL synchronization whereby the NW-TT can perform operations similar to that of DS-TT as in the DL synchronization, i.e., the DS-TT applies TS_i while the NW-TT applies TS_e. However, the difference in the UL synchronization is that instead of the source distributing timing information (UE in this case), UPF delivers gPTP messages to TSN end stations behind the DS-TT but without sending it back to the source UE.

B. TRAFFIC SHAPING AND SCHEDULING

The 5G capability of URLLC enables integration of TSN traffic shaping and scheduling. The URLLC service is provided through several components including integrated frame structure, flexible scheduling, efficient control and data resource sharing, and advanced channel coding schemes. URLLC is, however, possible due the development and implementation of 5G NR that supports sub-6 GHz and mmWave carrier frequency ranges. NR supports mini-slots comprising 2, 4, or 7 OFDM symbols whose transmission can start instantly with priority and preemption support. Such transmission capabilities facilitate URLLC and are beneficial for time-critical applications.

In the DL, a URLLC packet can follow either an instant or a reservation-based scheduling scheme [106]. Instant scheduling can interrupt an ongoing transmission via preemption indication sent by the gNB to the UE informing that the resources indicated contain URLLC data. In the reservation-based scheduling scheme, resources are reserved in advance for URLLC transmission [107]. For the UL, UE uses prior configured parameters through the radio resource control (RRC) signaling to periodically send UL grant free (GF) URLLC transmission. Successful reception of URLLC transmission is guaranteed by using one of the GF retransmission schemes [108].

5GS supports periodic deterministic communication both in DL and UL where QoS characteristics are known in-advance. The transmission schedule for both DL and UL, i.e., from the UPF to UE and vice versa, is provided externally to the 5GS such as via IEEE 802.1 TSN [95]. The knowledge of the traffic schedule is useful for 5GS since it allows the gNB to efficiently allocate resources using traffic scheduling schemes both for DL and UL transmission.

The 5G deterministic support features include TSC assistance information (TSCAI) and hold & forward buffering mechanism [95]. TSCAI describes the traffic flow pattern while hold & forward mechanism allows traffic flow de-jittering. The TSN AF interacts with the CNC to obtain TSCAI parameters. TSCAI parameters include traffic burst arrival time (latest possible time when the first packet of the data burst arrives), periodicity (time period between start of two bursts), and flow direction (UL or DL). The TSCAI is derived on a per QoS flow basis and is forwarded to the SMF via PCF (see Fig. 6). The SMF further signals the TSCAI parameters to 5G RAN with respect to the 5GS clock. The clock mapping for burst arrival time and periodicity from TSN to 5GS is the responsibility of SMF as it is measured and reported by the UPF.

TSN traffic arriving at the 5GS bridge through a corresponding PDU session is mapped with appropriate QoS flow configuration characteristics. The QoS flow between the UE and UPF can be characterized in terms of resource type, priority level, packet delay budget (PDB), packet error rate (PER), averaging window, and maximum data burst volume (MDBV). The resource types can be guaranteed bit rate (GBR), non-GBR, or delay critical GBR [11], [95]. Averaging window represents the duration over which the guaranteed flow bit rate (GFBR) and maximum flow bit rate (MFBR) are calculated. GFBR is the guaranteed bit rate to be provided to a QoS flow while MFBR limits the highest expected bit rate for the QoS flow over an averaging window.

Based on the QoS characteristics, 5G quality indicator (5QI) values have been standardized, however, this can also be pre-configured, or assigned dynamically. Delay-critical GBR resource type along with TSCAI parameters are used to define TSC QoS flows which may use either of the standardized, pre-configured, or dynamically assigned 5QIs. Nevertheless, to optimize the scheduling process, standardized 5QI values along with delay critical GBR can be used for industrial applications. The periodic TSC flow uses a burst size not exceeding MDBV within defined PDB. PDB is defined as an upper bound time that a packet may be delayed in 5GS. In other words, the largest amount of data to be transmitted for periodic TSC flow is mapped to a 5QI considering MDBV and PDB constraints.

As stated earlier, hold & forward buffering is another feature that 5GS supports for deterministic communication. This feature is identical to the gate scheduling behavior of the eight queues configured in the TSN GCL. Accordingly, frames of a corresponding queue in a given time interval are transmitted if the gate is open but are held back otherwise.

According to the 3GPP Release-16 specifications, VLAN configuration information for 5G TSN bridge is pre-configured with a mapping table at TSN AF and NW-TT. Moreover, the VLAN configuration information does not need to be exchanged between the TSN AF and NW-TT. The mapping table contains information regarding TSN traffic classes, UPF-UE delay, and priority levels. The PCF provides mapping information from TSN QoS to 5G QoS profile. PCF can trigger PDU session modification procedure for a new 5G QoS flow when initiated by the TSN AF depending up on the traffic requirements. Similar to the fixed TSN switches, solution for CNC controlled dynamic VLAN configuration for 5GS bridges is under consideration in 3GPP Release-17 specifications [98].

Allocation and retention priority (ARP) is the set of pre-configured values for TSC flows and contains information about the priority level, preemption capability, and preemption vulnerability. ARP, therefore, defines the importance of a QoS flow where the priority level ranges between 1 and 15, 1 being the highest. ARP can be used to decide which TSC flow may be accepted, rejected, or preempted, for example, in the case of resource limitations.

Although 5G specification provides comprehensive mapping to TSN traffic shaping and scheduling, the wireless nature that allows mobility and frequent changes in the network layout may require further enhancements [15]. Possibly, more QoS profiles specific to TSN communication are desirable, however, further evaluations of the 5GS TSN integrated system may provide future directions.

C. ULTRA RELIABILITY

TSN specifies frame replication and elimination for reliability (FRER), path control and reservation (PCR), and per-stream filtering and policing (PSFP) standards to achieve reliability. Reliability, however, depends on communication service availability that according to 3GPP ranges between 99.9999% and 99.999999% for industrial applications [101]. Reliability is required in both the control signaling and data transmission. It can be achieved through various techniques including HARQ, enhanced demodulation efficiency, multi-antenna transmission, use of multiple carriers, coordinated multipoint, and duplicate packet transmission over independent paths [14], [15]. The 3GPP Release-16 suggests an always-on PDU session for URLLC QoS flows. However, the responsibility lies with the UE to gain knowledge on whether the QoS flow is meant to serve URLLC traffic [95]. The 5G multipath transmission support increases reliability for URLLC traffic and enables TSN FRER but requires increased usage of resources [15]. Approaches to enable redundant transmission paths are discussed in the following.

1) REDUNDANT TRANSMISSION PATHS

The 5G system supports reliability by setting up two redundant PDU sessions over user planes as disjoint sessions (Fig. 6). Redundant PDU session establishment request is

initiated by the UE and is indicated to the SMF. Whether a UE can have a redundant PDU session is dependent upon its subscription. Multiple solutions to enable redundant transmission paths are suggested [95], [109]. These solutions are detailed as follows.

1) *Redundant user plane paths based on dual RAN connectivity*: In this approach, redundant user plane paths based on dual RAN connectivity are established (Fig. 8(a)). The establishment of two independent user plane paths for disjoint PDU sessions is dependent upon the availability and connectivity coverage of dual RAN (master and secondary) and UE's dual connectivity support. Once the RAN and UE dual connectivity support is available, the TSN FRER protocol is responsible to manage replication and elimination of redundant frames over duplicate paths. A UE connects to both the master and secondary RAN, however, the master RAN controls the selection of secondary RAN and enables the set up of dual connectivity feature via the Xn interface.

2) *Multiple UEs per end station for user plane redundancy*: In this approach, the end station is equipped with two UEs where each UE can set up an independent PDU session via separate paths connecting to RAN 1 and RAN 2, respectively (Fig. 8(b)). Again, it is assumed that multiple RAN nodes can provide coverage to the UEs equipped by the end station. The two PDU sessions that establish via UPF 1 and UPF 2 connect to the same TSN network and are controlled through their respective SMFs. To ensure that the UEs connect to different RANs, the operating frequency of the RAN nodes can be configured distinctly. Alternatively, separate reliability groups for the UEs and cells of the RAN can be defined to allow connectivity of the UEs to different RAN nodes. However, the redundant UE approach does not seem suitable for highly mobile end stations.

3) *Redundant transmission with two N3 tunnels between the user plane and RAN*: Another approach to support high reliability is through a redundant transmission link between the RAN and UPF. This link is referred to as the N3 tunnel. In this approach reliability is assumed at the RAN, UPF, and control plane network functions for URLLC traffic. However, the N3 tunnel is considered unreliable, possibly due to the environmental factors that affect the communication at the backhaul network. In order to enhance reliability between the RAN and UPF, two N3 tunnels can be provisioned over disjoint transport layer paths associated with a single PDU session. Different routing information such as the use of different IP addresses or different network instances can be provided by the UPF or SMF for routing the traffic over disjoint paths. Once the redundant paths are provisioned, the SMF indicates the RAN and UPF about the redundant tunnel belonging to the same PDU session. Replicated packets are assigned the same GPRS tunnelling protocol user plane (GTP-U) sequence number and are transmitted via the N3 Tunnel 1 and the N3 Tunnel 2 separately (Fig. 8(c)). In the DL, the packet received first by the RAN from either of the tunnel is forwarded to the UE while the replicated packet is dropped. In the reverse direction, packets are replicated by the RAN that assigns the

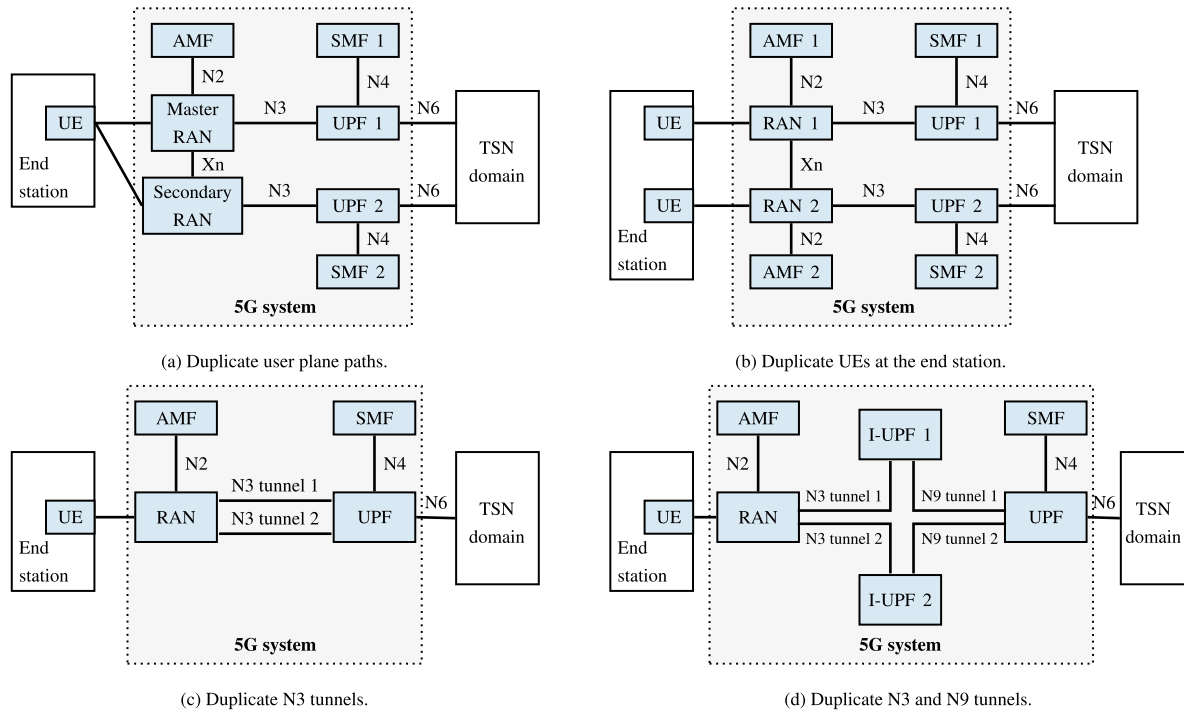


FIGURE 8. Approaches to enable redundant transmission paths through duplication of transmission chains [95], [109].

same GTP-U sequence number and are forwarded to the UPF using separate N3 tunnels.

4) *Two N3 and N9 tunnels between RAN user plane:* The redundant N3/N9 tunnel is an extension of the two N3 tunnel approach where two intermediate UPF (I-UPF) are added to support redundant URLLC transmission between the UPF and RAN. The N3/N9 redundant transmission is performed at QoS flow granularity that shares the same QoS flow ID of the same PDU session. Similar to the redundant N3 tunnel approach, in the DL, the UPF connected to the TSN domain duplicates the packet and assigns them the same GTP-U sequence number. The duplicated packets are transmitted to I-UPF 1 and I-UPF 2 separately and are forwarded to the RAN via the N3 Tunnel 1 and N3 Tunnel 2, respectively (Fig. 8(d)). The duplicate received packet is eliminated by the RAN based on the GTP-U sequence number. In the UL, the URLLC packet received from the UE is duplicated by the RAN that follows the two N3/N9 redundant paths and the UPF eliminates the duplicate received packet.

2) 5G SUPPORT FOR PSFP

The 5GS support for PSFP depends upon the PSFP implementation support from TT functions on the device side and the network side. The TSN AF interfaces towards the CNC to obtain information on PSFP objects corresponding to the PSFP functionality implemented by the DS-TT and NW-TT. The TSN AF extracts relevant PSFP configuration parameters provided by the CNC to determine traffic pattern information. Stream parameters such as the maximum number of stream filter instances, the maximum number of stream gate instances,

and the maximum number of flow meter instances are extracted. In case PSFP information is not available from the CNC, pre-configured QoS flows are used and configured during the PDU session establishment. If the PSFP information is available, the TSN AF uses PSFP information to determine the traffic pattern information such as traffic periodicity, burst arrival time, burst size, and maximum flow bitrate of a TSN stream. Moreover, the TSN AF identifies traffic flow direction (DL or UL), traffic class of the TSN stream with their per port priorities, and transmission gate scheduling parameters. The TSN AF is responsible to forward stream parameters, TSCAI (flow direction, periodicity, and burst arrival time), priorities, gate scheduling, and other traffic pattern information in a TSC assistance container to the SMF via PCF (see Fig. 6). The mapping of the traffic class to TSN QoS and delay requirements can be performed using TSN AF QoS mapping table. Hence, the corresponding 5G QoS flow can be configured by an appropriate selection of 5QI in the PCF. If, however, the standard or pre-configured 5QIs do not satisfy the required TSN flow QoS, the SMF can reconfigure the QoS flow binding. TSN streams that terminate to the same egress port and correspond to the same traffic class with the same periodicity and burst arrival time may aggregate to the same QoS flow. The PCF derives the required MDBV taking the maximum burst size as input of the aggregated TSN streams.

D. RESOURCE MANAGEMENT

5G supports DL and UL resource reservation schemes for URLLC traffic as discussed in Section VI-B. 5G also supports TSN integration as a logical bridge to the TSN domain on a

per UPF basis. A single or multiple 5G TSN logical bridge may be available to the TSN domain whereby each logical bridge is bound by a UPF ID [110]. A 5G TSN logical bridge consists of ports both on the UPF side (NW-TT) and the UE side (DS-TT) as well as the UE-UPF tunnel (see Fig. 6). The UPF side ports support connectivity to the TSN domain via NW-TT, while the DS-TT side port(s) are associated with the PDU session provided connectivity to TSN enabled devices. The binding relationship between the DS-TT and the PDU session is stored at the TSN AF. The NW-TT and DS-TT port capabilities such as the UE-DS-TT residence time and bridge propagation delay are notified to the CNC through the TSN AF for registration or modification [95], [110]. The same UE-DS-TT residence time is used for both UL and DL and applies to all traffic classes. However, the residence time may vary among the UEs. The TSN AF interacts with the CNC to provide port capabilities and delay information which enables the CNC to schedule traffic accordingly [67].

The TSN AF determines the required QoS for one or more TSN streams based on the information received from the CNC (e.g., GCL), bridge delay, and UE-DS-TT residence time. It forwards the requirement request to the PCF. The PCF maps the TSN QoS parameters with 5G QoS and notifies them to the SMF. The SMF establishes/modifies the QoS flow of a PDU session for the TSN stream(s) and informs the UPF regarding necessary modifications. The SMF determines the TSCAI for the requested QoS flow and sends them to the RAN together with the QoS profile for resource reservation.

The 5G grant-free scheduling to support URLLC traffic provides reliability and low latency for periodic data transmission. However, aperiodic or sporadic traffic may not receive guaranteed high reliability with bounded latency requirements due to the impact of self-collisions [111]. Solutions addressing grant-free scheduling for aperiodic traffic to meet URLLC requirements may be required in the future 3GPP releases.

E. NETWORK CONFIGURATION

5G supports TSN centralized architecture where the TSN AF interacts with the CNC for scheduling and identifying QoS requirements [11]. The fully distributed TSN configuration and the hybrid models are yet not supported by the 5GS. A direct connection of the 5G user plane with the TSN enabled switches can be envisioned for distributed and hybrid models due to the absence of CNC [15]. For these two models, distributed resource reservation protocol namely, P802.1Qdd - resource allocation protocol (RAP) is under development. 5G integration for distributed model shall be expected after the development of RAP [67].

F. EXAMPLE 5G-TSN INTEGRATED SCENARIO

In an industrial environment, an example scenario widely realized is a closed loop application such as the controller-to-controller (C2C) communication where a control entity communicates with another control entity of a machine [67]. As a 5G integrated TSN system, a machine controller can be connected to the wired TSN backbone network while a

second controller can be connected to the 5GS via a UE. Therefore, the two controllers connect via the 5G integrated TSN system, one via the NW-TT port (at the UPF) and the other via the DS-TT port (at the UE). Such a C2C use case typically requires communicating cyclic synchronous traffic. The 5GS system supports such a communication by establishing one or multiple PDU sessions whereby the configuration on the QoS requirements is communicated and orchestrated via the TSN AF. Although, the communication requirements depend on the application itself, the 5GS supports TSN standards of IEEE 802.1AS, IEEE 802.1Q, IEEE 802.1Qbv, IEEE 802.1CB, IEEE 802.1Qci, and IEEE 802.1Qcc that may be required to support such a C2C use case.

VII. USE CASES

Wireless TSN enables flexibility, mobility, easier network re-configuration, and lower maintenance costs along with high reliability and deterministic communications [17], [67]. Several use cases could benefit from the capabilities of the wireless TSN technology. Some of these use cases are described below and summarized in Table 2 along with their requirements.

A. INDUSTRIAL AUTOMATION

A number of use cases for industrial automation can be identified where wireless TSN will make a significant impact. These may include closed-loop control, HMI and production IT, and functional safety to mention a few. The closed-loop control include very low latency, tight synchronization, and minimum jitter [101]. The process monitoring use case employs multiple sensors to collect and analyze data. The data collected can be used for predictive maintenance, processes optimization, or condition monitoring [112]. Here time synchronization is a requirement for consistent and accurate monitoring of data. In an industrial setting different HMIs may exist including interface panels to different machines and IT devices apart from production IT (e.g., enterprise resource planning (ERP) and manufacturing execution system (MES)) [113]. Lastly, functional safety applications include fast motion control, emergency-stop, and overhead cranes, etc. They impose very short response time to ensure human safety and so low latency and ultra-high reliability is required for such systems [113], [114]. Both 802.11 and 5G technologies may be suitable for industrial automation use cases in general, though the specific characteristics of each scenario must be considered (e.g., the required coverage area).

B. AUTOMOTIVE

The number of sensors in a car are increasing exponentially given the new advanced driver assistance systems (ADAS) and autonomous driving (AD) functions along with state-of-the-art infotainment systems. Different types of sensors and actuators are used including environmental monitoring sensors, cameras, ultrasonic, light detection and rangings (LIDARs), and radio detection and rangings (RADARs). Sensor

TABLE 2. Wireless TSN Use Cases With Requirements

Domain	Use case	QoS Requirements		
		Latency (ms)	Reliability (%)	Throughput (Mbps)
Industrial automation	Closed loop control	0.5-1	99,999-99,99999	0.1-5
	Process monitoring	4-8	99,99-99,9999	5-10
	HMI and production IT	50-200	99,9-99,99	1-2
	Functional safety	0.5-1	99,9999-99,99999	0.1-5
Automotive	Autonomous driving	1-10	99,999-99,99999	>1
	In-Vehicle networking	2-10	99,99-99,9999	>2
Audiovisual	Cloud gaming	5-50	>99.9	10-35
	Interactive museum	10-20	99.9	25-500
Aerospace	Space shuttle	1-10	99,9999-99,99999	50-100
	Aircraft	<20	99,9999	100
	UAVs	20-200	99,9-99,99	0.03-120
Railway	Backbone network	16-500	99,9999	100
	Consist network	4-250	99,9999	100
Healthcare	Assisted living	100-500	99,9-99,999	5-10

data, audio/video transmission, and control messages present different latency and reliability requirements to the communication network. The automotive industry can benefit from the features of TSN [115] and some TSN solutions for the automotive industry are already available on the market [116], [117]. Nevertheless, a wireless TSN removes all the wires, which simplifies communication network inside the vehicle and consequently the vehicle design procedure [118]. Clearly, 802.11 technology for enabling wireless TSN is more suitable for in-vehicle communications because of its limited coverage, whereas 5G technology is more suitable for inter-vehicle and vehicle to infrastructure communications focusing AD.

C. RAILWAY INDUSTRY

The European railway industry has positioned to be a key player in the transport sector in the upcoming years. To this aim, the Shift2Rail Multi-Annual Action Plan has identified several challenges that need to be tackled. One of them is the evolution of the current wired train control and monitoring system (TCMS) to a wireless architecture [119]. TCMS network operates in a two-level network architecture that comprises the train backbone, which connects different consist networks. The next-generation TCMS will apply wireless technologies at both backbone and consist levels, thus implementing wireless train backbone (WLTB) and wireless consist network (WLCN) solutions. These networks present diverse and stringent real-time requirements [120]. TSN has been chosen as a perfect candidate for the wired domain network [121], whereas both 5G and 802.11 have been chosen as the convenient solutions for WLTB and WLCN respectively.

D. AUDIOVISUAL

Audiovisual use cases incorporate both visual and audio interactions to create a real-time virtual environment. It requires high data rates for video transmission, low latency, and accurate time synchronization to avoid jitter in the audio and

video data. Several applications are under this use case. For instance cloud gaming is becoming popular and requires high traffic rate to ensure appropriate video resolution, low latency in the range of few ms and even lower jitter to ensure adequate user experience. In particular, 5G could be interesting to enable high-end graphics over hand-held devices with minimum consumption power. Other applications, such as augmented reality (AR) and virtual reality (VR), are also emerging audiovisual trends. On the VR side, some VR glasses rely on external processing devices to compute the graphics. Given that the state-of-the-art glasses have a refresh rate of 120 frames per second and 2 k resolution, their cycle time is in the range of 8.3 ms, whereas the throughput is around 200 Mbps. The distance between the VR glasses and the computer is small and so 802.11-based TSN is a suitable choice for this use case. On the AR side, multi-user AR games require fast updates of the AR data represented to each user. As a result, their latency requirement is in line with VR, though it requires far lower throughput as it only sends the status of each player and the AR elements. 5G may be more suitable because multi-user AR may be deployed in large open spaces.

E. AEROSPACE

Communication within an aircraft consists of different traffic classes from time-critical to best-effort traffic. Hence correct scheduling according to communication constraints is important. TSN has been envisioned as a cabin backbone bus for cabin communication in aircrafts' [122] or in NASA Orion project for the multi-purpose crew vehicle [123]. Moving to a wireless TSN can be desirable for the above use cases as it would not only reduce the weight of the vehicle by removing wires but also lower the maintenance and reconfiguration costs. Some wireless solutions for in-cabin communication in aircraft and spacecraft have already been considered [124]–[126].

Another consideration under aerospace is of small unmanned aerial vehicles (UAVs). UAVs are being used for many commercial applications and the communication requirements vary depending upon the application scenario. Some of these applications such as construction, search & rescue, and entertainment require multi-UAV collaboration where synchronization is needed [127]–[131]. A 5G supported TSN service could be a suitable choice to enable unprecedented levels of coordination between UAVs.

F. HEALTHCARE

According to the 2017 UN report “World Population Aging” [132], the number of 80 years old or above people will increase threefold from 2017 to 2050. This increase in the aging population and developments in non-obtrusive and embedded sensor systems are changing the way the medical staff provides care to the elderly. The possibility for remote and continuous physiological-monitoring ensures timely intervention from a medical practitioner. Additionally, care can be provided with the help of health care robots for the disabled and cognitively impaired population [133], [134]. The physiological-monitoring systems need to transmit video/audio along with sensor data with high reliability and in emergency cases with low latency. Additional requirements include flexibility and support for mobility. Wireless TSN can meet these requirements in order to realize the evolving assisted living use cases.

VIII. CONCLUSION

In this paper we present the standardization activities and developments in 802.11 and 5G to enable wireless TSN. In particular, we focus on analyzing 1) the enabling mechanisms in both 802.11 and 5G for time synchronization, traffic shaping and scheduling, ultra-reliability, resource management, and flexible network configuration models and 2) the current state of the integration of both 802.11 and 5G with wired TSN. Additionally, we present wireless TSN use cases in different domains along with their QoS requirements in terms of latency, reliability and throughput.

Currently, wireless TSN is in its infancy, and there still are many open questions till we can reach open standardization. The 802.11 working group has been considering new mechanisms to enable wireless TSN like operation over 802.11. Some existing mechanisms already enable few TSN features over 802.11ax (e.g., time synchronization) though several open questions are still to be solved for the next standard release. The probability of interfering transmissions is by far the most serious hurdle to achieve ultra-high reliability and guaranteed latency, especially for communications in unlicensed bands. However, even if 802.11be specification is able to solve these challenges, its adoption will strongly depend on the specific implementations of chip makers and in the integration model between 802.11 and wired TSN. Therefore, the most critical and often overlooked challenge is the definition of the mechanisms to enable the seamless integration of wired TSN and 802.11be. Regarding 5G, the open questions among

others include the integration of TSN configuration and management with 5G management and orchestration capabilities, TSN domain synchronization through 5G internal clock, and the integration of TSN distributed and hybrid models with 5G as the standards become available. Moreover, apart from of 5G TSN integration, another research direction could be provisioning of deterministic communications within 5G which would require modifications in the core and RAN entities of the 5GS.

Nevertheless, the developments of wireless TSN are gaining interest and momentum among researchers, the question is when will it be realized for industrial use cases. Certainly, newer challenges will then emerge.

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